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Background

URDF: Unified Robot Description Format,
in XML format

URDF bundle example:

```
kuka_kr3_support/
├── urdf/
│   └── kr3r540.urdf
└── meshes/
    ├── visual/
    │   ├── link_0.dae
    │   ├── link_1.dae
    │   └── ...
    └── collision/
        ├── link_0.stl
        ├── link_1.stl
        └── ...
```

Research Questions

Q1 (Sim-to-Sim):

Do different simulators render the same manipulator URDF bundle in the same way?

Q2 (Real-to-Sim):

Given a real robot, is there a way to find corresponding URDF bundle from its appearance?

Broad Impact:

Bridge Sim-to-Sim:

- Support the reuse of simulated robot data,
- Improve the transferability of robot learning models trained on simulated data

Bridge Real-to-Sim:

- Ease the development of URDF bundle for new manipulators.

Canonical-View URDF Dataset

- Link: <https://github.com/BotYue/manipURDF4view>
- 131 manipulator URDF bundles rendered in both MuJoCo and PyBullet.
- Four views per robot: isometric, front, top, side.



• Dataset stats:

Manufacturer	Count	Source Repository	Count
FANUC	42	ROS-Industrial	86
ABB	23	MATLAB Robotics	15
KUKA	19	Robotics Toolbox	13
Universal Robots	11	Drake	7
Stäubli	10	Others	10
Trossen Robotics	9		
Others	17		

Dataset QR:



Sim-to-Sim Matching

- Different simulators can render the same URDF bundle differently, due to mesh file interpretation, camera setup, lighting, shadowing, ...



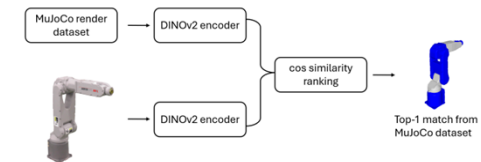
Sim-to-Sim Matching (cont.)

- A visual matching test is conducted:
using MuJoCo renderings as queries
and PyBullet renderings as the retrieval library.
- 6 methods used to compare visual similarity.
CLIP, I-JEPA, DINOv2 and SigLIP2 use pretrained models to get image embeddings.
- Matching results:
present both single-view (isometric only) similarity and four-view equal-weighted similarity

Method	Iso-Only		4-View	
	Top-1	Top-5	Top-1	Top-5
Pixel MSE	7.6%	18.3%	8.4%	18.3%
HOG	9.2%	43.5%	17.6%	48.1%
CLIP	22.9%	50.4%	36.6%	63.4%
I-JEPA	29.8%	67.2%	38.2%	77.1%
DINOv2	42.0%	79.4%	61.8%	93.1%
SigLIP2	23.7%	61.8%	49.6%	90.1%

Real-to-Sim Matching

- Propose a real-to-sim matching pipeline.
- A real photo of a manipulator is used as the query.
The retrieval library consists of MuJoCo isometric-view renderings of all manipulator bundles in the dataset.



Note: The extended work will be presented at AAAI Fall Symposium 26 - Unifying Representations for Robot Application Development

References

- [1] D. Tola and P. Corke, "Understanding URDF: A dataset and analysis," RA-Letters, 2024.
- [2] D. Tola and P. Corke, "Understanding URDF: A survey based on user experience," IEEE CASE, 2023.